

Clock-Valid pattern detection Functional Design Specifications

UCIe 3.0 physical layer

Version 1

Block Owner

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2 Revision History

Version	Date	Author(s)	Revision Notes	Owner Approval

3 Overview

This module detects whether incoming clock and valid signals conform to a known alternating pattern. It supports two operating rate modes — half-rate and quadrature-rate — and two detection algorithms: a per-lane (consecutive-correct) method and a compare/error-count method.

4 Operation and Description

4.1 Digital Interface

4.1.1 Parameters Names

Parameter Name	Default	Description
p_CLK_SEQ	48'hAAAA_AAAA_0000	The correct sequence of the clock pattern

4.1.2 Ports Names

Port Name	Port Width	Port Type	Description
i_clk_p	1	Input	Clock input from the phy-layer
i_clk_n	1	Input	Clock input from the phy-layer with different phase
i_valid	1	Input	Valid signal under test.
i_track	1	Input	Track input from the phy-layer with different phase
i_hclk	1	Input	Half-rate reference clock.
i_dclk	1	Input	Full-rate reference clock.
i_reset	1	Input	Asynchronous active-high reset.
i_halfrate	1	Input	Rate mode selector. 1 routes half-rate detector outputs to o_valid_result and o_clk_result; 0 routes quadrature-rate detector outputs.
i_pattern_type	1	Input	Detection enable flags. Bit [1] enables valid detection; bit [0] enables clock detection.
i_detection_type	2	Input	Valid detection algorithm selector. 0 selects per-lane (consecutive-correct) mode; 1 selects compare/error-count mode
i_error_threshold	16	Input	Error count threshold for compare-mode valid detection. The compare result de-asserts when the cumulative mismatch count exceeds this value
o_clk_result	1	Output	Clock pattern detection result.
o_valid_result	1	Output	Valid pattern detection result

4.2 Functional Description

1. Valid Pattern Detection

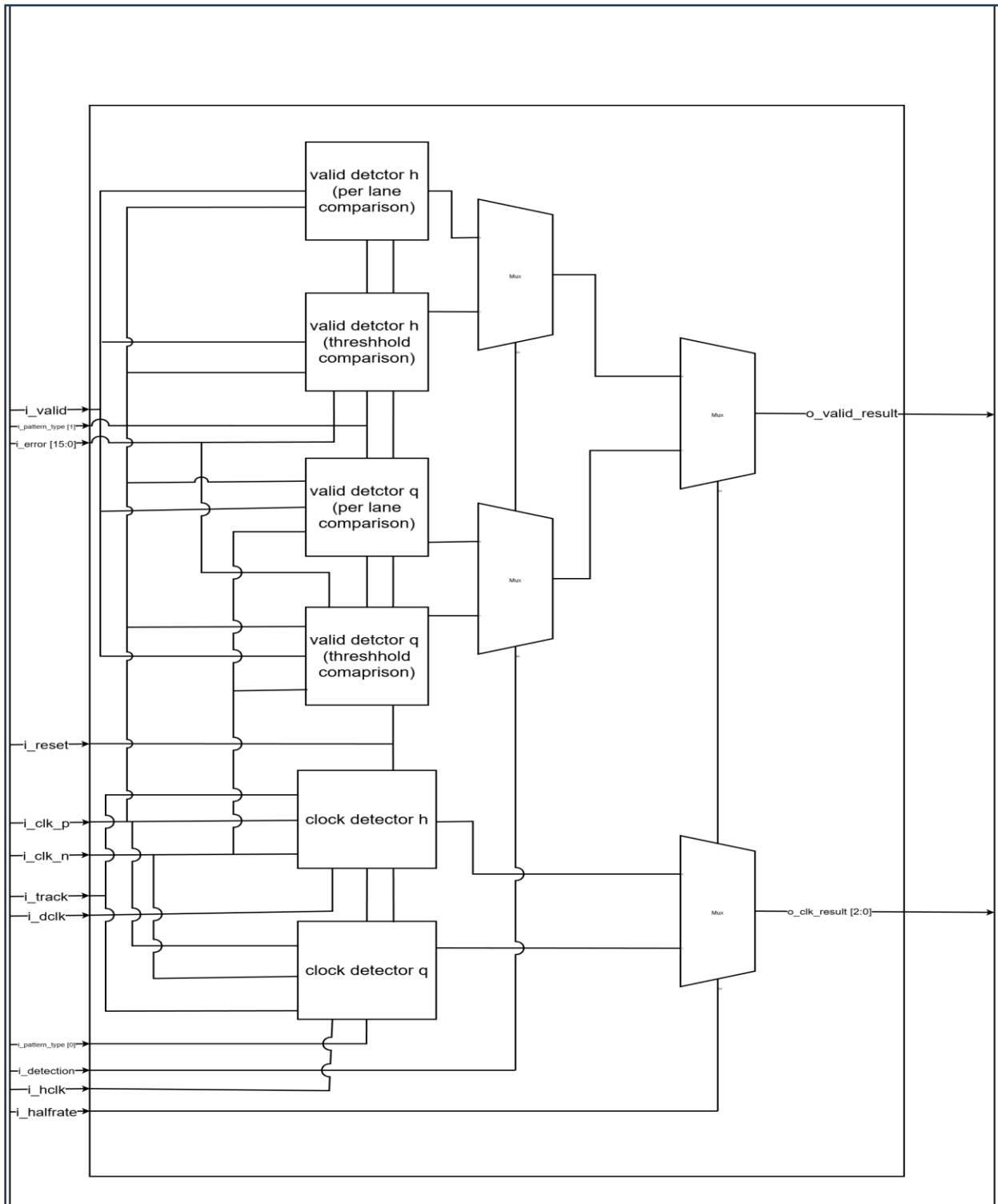
- **Half-Rate Mode**
 - The `i_valid` signal is sampled on the rising edge of a **gated `i_clk_p`**. Every 8 samples, the 8-bit shift register is compared to the expected pattern `8'b11110000`.
 - **Per-Lane Detector:**
 - A clock gate disables sampling once the result asserts (self-stopping).
 - A 4-bit `counter_correct` tracks how many consecutive 8-sample windows matched.
 - `w_valid_result_per_lane_h` output of valid detector h (per lane comparison) asserts when 16 consecutive windows match.
 - **Compare/Error-Count Detector :**
 - On every 8-sample window, a mismatch increments a 16-bit error counter.
 - `w_valid_result_compare_h` stays high as long as `counter_error_h` ≤ `i_error_threshold`.
 - Sampling ends after deasserts when the counter exceeds the error threshold.
 - A local mux selects between the two results based on `i_detection_type`:
 - 0 → per-lane result
 - 1 → error-count result
- **Quadrature-Rate Mode**
 - Identical in algorithm to half-rate, but `i_valid` is sampled on **both rising edges of `i_clk_p` and `i_clk_n`**, effectively doubling the sampling rate. Two independent clock gates are maintained — one for each differential clock edge — both driven by the same enable logic.
- **Active Mode**
 - **Enable Condition**
 - Detection is gated by `i_pattern_type`. If the pattern type is not `2'b11`, all shift registers and counters are cleared and detection is suppressed.
 - **Sampling & Serialization**
 - The **H channel** samples `i_valid` on **both edges of `i_clk_p`** (single-edge clocking).
 - The **Q channel** samples `i_valid` on **both edges of `i_clk_p` and `i_clk_n`** (double-edge / quadrature clocking).
 - In both channels, each incoming `i_valid` sample is shifted into an 8-bit shift register.

- **Start Condition**
 - Detection begins when i_valid is first asserted while the channel enable flag is low. On this event, the enable flag is set and shifting begins.
- **Pattern Check (Every 8 Samples)**
 - When the sample counter reaches **8**, the accumulated shift register content is compared against the expected pattern 8'b11110000:
 - **Match:** The valid result (w_valid_result_active_h/q) retains its asserted state (1).
 - **Mismatch:** The valid result is **de-asserted (0)**, indicating an invalid or unexpected pattern.
 - After the check, the counter is reset based on the current state of i_valid:
 - If i_valid is still high → counter resets to 1 (carry-over of the current sample).
 - If i_valid is low → counter resets to 0 and the enable flag is cleared, returning to idle.

2. Clock Pattern Detection

- **Half-Rate Mode**
 - Uses a gated i_dclk (full-rate reference clock) as the sampling clock, sampled on the negative edge.
 - Capture is armed on the first detected high of i_clk_p to align phase. If this is wrong whatever the pattern comes it will be detected wrong because the phase is not aligned. (this is limited because it is an analog model)
 - Three 48-bit shift registers accumulate i_clk_p, i_clk_n, and i_track in parallel.
 - Every 48 samples, each shift register is evaluated independently:
 - i_clk_p must match CLK_SEQ
 - i_clk_n must match ~CLK_SEQ (inverted — differential pair relationship)
 - i_track must match CLK_SEQ
 - Each signal has its own 4-bit counter_correct. A result bit asserts after 16 consecutive matching 48-bit windows.
 - The clock gate disables i_dclk once all three result bits are high (&w_clk_result_h).
- **Quadrature-Rate Mode**
 - Uses a gated i_hclk (half-rate reference clock).
 - i_clk_p and i_track are serialized on the negative edge of i_hclk; i_clk_n is serialized on the positive edge — using two separate always blocks with independent counters and arm/enable flags.
 - Unlike half-rate, i_clk_n is compared to CLK_SEQ (not inverted), because the quadrature relationship means it arrives 90° shifted rather than 180° inverted.
 - Each result bit follows the same 16-consecutive-window rule.
 - The clock gate disables i_hclk once all three bits are asserted.

4.3 Block Diagram



Note the valid detector in active mode is implemented entirely in the valid detector per lane